# Expectations vs Reality: AI for Learning

Group 13

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## 1 Introduction

### 1.1 Group Info

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## 2 Project Overview

People have many grievances about AI when used for learning: wrong outputs, hallucinations, and lengthy gibberish are among a few [2]. We decided to ask people about common problems they face when using AI (particularly LLMs) in educational settings [1] and then Benchmark the Most commonly used models against AMC Maths Problems [4]. Doing so allows us to investigate if there exists a gap between actual benchmark results and users' predicted results [3].

## 3 Methodology

We collected responses about common AI faults via a Microsoft form and a separate form was made to test the LLM's. The LLM form was based on AMC maths problems (the kind used in actual AI benchmarks) that tested logical and spatial reasoning in a variety of ways. The "human" form was used to gauge peoples methodology when using AI, their most frequently used models and what issues they faced. It finally ended with a very simple question that anyone should have been able to solve. However, we weren't looking for the answer, we were testing to see if people were willing to solve the question at all.

# 4 Key Findings

## 4.1 User Expectations: Human Form Analysis

Based on the results of the Human form, we were able to conclude that people used the following models the most:

| Name | n | Count % of answered (n) |
|---|---|---|
| ChatGPT | 31 | 66.0% |
| Gemini | 24 | 51.1% |
| Claude AI | 13 | 27.7% |
| Meta AI | 1 | 2.1% |

Table 1: Model Response Analysis

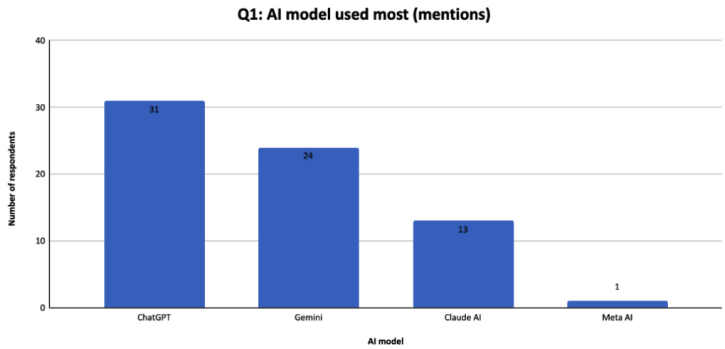



Figure 1: Most used model frequency count

Using the Equation:

$$\text{Model Percentage Use} = \frac{\text{Model Response Count}}{47} \times 100 \tag{1}$$

We calculated the percentage of respondents who used each model which helped us decide what niche and most used models to target.

Particularly Deepseek which none of the respondents seemed to use as can be seen in **Table 1** and **Figure 1**

## 4.2 LLM Benchmark Analysis

In contrast, we tested and then calculated the average score for the LLM Benchmark form which totaled to an impressive 5.8/6 across the 5 tested models:

Using the model Percentages (See Equation 1 for details.)

We bench marked the models against 7 AMC Maths problems which resulted in the following Average performance/scores:

Table 2: Model Rankings & Categorization

| Model Name | Model Tier | Score (Max 7) | Accuracy Rate | Completion Time (s) |
|---|---|---|---|---|
| GPT 6.1 Sol | Frontier / High-Reasoning | 7 | 100.0% | 87 |
| Deepseek V4.1 Flash | Frontier / High-Reasoning | 7 | 100.0% | 223 |
| Gemini 3.8 Flash | Mid-Tier / Generalist | 6 | 85.7% | 126 |
| Claude Sonnet 5.5 | Mid-Tier / Generalist | 5 | 71.4% | 81 |
| Perplexity Sonar Pro | Search / Specialized | 4 | 57.1% | 83 |
| **Average / Total** | — | **5.8** | **82.9%** | **120.0** |

# 5 Conclusion and Recommendations

In conclusion, there is an obvious gap between the actual AI output and what people expect from it. This could be improved by targeting niche models (like Deepseek which performed the best) and educating the average user about prompt engineering techniques.

# References


[1] Li Chen, Pingping Chen, and Zhiyong Lin. Artificial intelligence in education: A review. *IEEE Access*, 8:75264–75278, 2020.

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[4] Berkeley EECS Technical Reports. Benchmarking large language models on advanced mathematical reasoning. Technical Report EECS-2025-121, University of California, Berkeley, 2025.